

# Finding Zeros with the Rational Root Theorem

## Algebra 2

---

Testing candidates, factoring completely, and finding every zero. For a polynomial with integer coefficients, every rational zero has the form  $\pm \frac{p}{q}$ , where  $p$  divides the constant term and  $q$  divides the leading coefficient. The routine is always the same: **(1)** list the candidates, **(2)** test them until one gives a remainder of 0, **(3)** divide it out, **(4)** finish the depressed quotient by factoring or by the quadratic formula. Problems 1–2 walk that routine on a single polynomial; after that you run it yourself. Show your candidate list and your division work — both are part of the answer.

**1.** Let  $f(x) = x^3 - 4x^2 + x + 6$ . The rational root candidates are  $\pm 1, \pm 2, \pm 3, \pm 6$ . Evaluate  $f(2)$  to decide whether  $x = 2$  is a zero of  $f$ .

Answer: \_\_\_\_\_

**2.** Using your result from problem 1, divide out the corresponding factor and find *all* zeros of  $f(x) = x^3 - 4x^2 + x + 6$ .

Answer: \_\_\_\_\_

3. Factor  $g(x) = x^3 + 2x^2 - 5x - 6$  completely over the integers.

Answer: \_\_\_\_\_

4. Let  $h(x) = 2x^3 - x^2 - 8x + 4$ . Because the leading coefficient is 2, the candidate list includes halves. Evaluate  $h(\frac{1}{2})$  to decide whether  $x = \frac{1}{2}$  is a zero.

Answer: \_\_\_\_\_

5. Find all zeros of  $h(x) = 2x^3 - x^2 - 8x + 4$ .

Answer: \_\_\_\_\_

**6.** Find all zeros of  $p(x) = x^3 - 4x^2 + 2x + 4$ . Exactly one zero is rational — find it first, then solve the quadratic that is left. Give exact answers, not decimals.

Answer: \_\_\_\_\_

**7.** Factor  $q(x) = 3x^3 - x^2 - 12x + 4$  completely over the integers.

Answer: \_\_\_\_\_

8. Solve  $x^3 - x^2 - 13x - 3 = 0$ . Give all real solutions in exact form.

Answer: \_\_\_\_\_

9. Find all zeros of  $r(x) = x^4 - 2x^3 - 7x^2 + 8x + 12$ . A quartic needs the routine twice: after the first division you are left with a cubic, so run the theorem again on it.

Answer: \_\_\_\_\_

**10.** Find all zeros of  $s(x) = x^3 - x^2 + 4x - 4$ , including any non-real zeros. State how many zeros the theorem itself was able to supply.

Answer: \_\_\_\_\_

**11.** Factor  $t(x) = 4x^3 - 8x^2 - x + 2$  completely over the integers.

Answer: \_\_\_\_\_

**12. Synthesis.** For  $u(x) = x^3 - 3x - 1$  the candidate list is just  $x = 1$  and  $x = -1$ , and  $u(1) = -3$  and  $u(-1) = 1$ , so neither is a zero. A classmate concludes: “The rational root theorem found nothing, so  $u$  has no real zeros.” Explain in two or three sentences why that conclusion is wrong, and give evidence that  $u$  does have a real zero between  $x = -1$  and  $x = 1$ .